GRIDINTEL · MINI-GRID INTELLIGENCE

Mini-grid compliance and results

Availability and hours of supply, connections, the tariff as an assumption, and each site's report against its records.

SAMPLE: SYNTHETIC DATA

This report is a sample. It shows the format and the methods GridIntel would deliver; it is not a report of real results. Every figure in it is calculated from a designed demonstration dataset, and none describes a real network, site, customer or organization.

A designed demonstration dataset for September 2026. It describes no real network, customer, meter reading, outage, bill or payment.

FOR
Mini-grid regulator; programme funder

RECORDS OF
Demonstration Mini-grid Operator (synthetic)

PERIOD
14 September 2026

REPORT ID
GI-MGC-2026-09-14-0B710C33

FIGURES COMPUTED
20 Oct 2026, 10:10 WAT, from data as of 15 Sep 2026, 00:00 WAT

DEFINITION
Version 1

DATA FINGERPRINT (SHA-256)
b0167ca08ac57457af3bea882205afb118f40ea88543b10d1bbe5fea52b64234

IN BRIEF

SITES IN THE REGISTER
4 • C
OK · Calculated

AVAILABILITY, DRY RIVER
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

AVAILABILITY, GRANITE HILL
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

AVAILABILITY, PALM LANDING
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

AVAILABILITY, TAMARIND CROSS
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

Tariffs: Tariffs are assumptions, not current published rates.

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1 Supply: availability and hours of supply

Whether each site's customers had power, and for how many hours a day.

Availability and hours of supply, by site

Availability is customer-weighted, with every recorded interruption counted; it is shown again with planned interruptions left out.

	AVAILABILITY	WITHOUT PLANNED	HOURS OF SUPPLY A DAY	INTERRUPTIONS PER CONNECTION
Dry River Commissioned 2019-11-01	100.0% • C	100.0% • C	24.0 h/day • C	0.0 • C
Granite Hill Commissioned 2024-09-01	100.0% • C	100.0% • C	24.0 h/day • C	0.0 • C
Palm Landing Commissioned 2021-06-01	100.0% • C	100.0% • C	24.0 h/day • C	0.0 • C
Tamarind Cross Commissioned 2023-03-01	100.0% • C	100.0% • C	24.0 h/day • C	0.0 • C

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

Interruptions

Every interruption recorded at a site in the period, by site and in order.

> No interruption is recorded at any site for 14 September 2026.

2 Connections and consumption

How many customers each site connects, how many bought or used energy, and how much.

Connections, by site

Connections in the register, and those with an active account. Per-connection figures are over active connections. Consumption is from customer meters; energy sold is energy billed.

	CONNECTIONS	ACTIVE	ENERGY SOLD PER CONNECTION	CONSUMPTION PER CONNECTION	CUSTOMER METERS
Dry River 45 kWp PV · 90 kWh battery · 60 kVA diesel generator	300 • C	300 • C	0.9 kWh • C	*insufficient data* !	294 of 300 recorded for the whole period; 0 unmetered; 6 incomplete
Granite Hill 60 kWp PV · 150 kWh battery · no generator	130 • C	130 • C	0.2 kWh • C	0.4 kWh • C	130 of 130 recorded for the whole period; 0 unmetered; 0 incomplete
Palm Landing 80 kWp PV · 200 kWh battery · 40 kVA diesel generator	417 • C	393 • C	0.3 kWh • C	0.7 kWh • C	417 of 417 recorded for the whole period; 0 unmetered; 0 incomplete
Tamarind Cross 100 kWp PV · 240 kWh battery · 50 kVA diesel generator	465 • C	465 • C	0.7 kWh • C	0.8 kWh • C	465 of 465 recorded for the whole period; 0 unmetered; 0 incomplete

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

3 Generation

How much of each site's energy was renewable, and how its plant performed.

Renewable fraction and plant, by site

Renewable fraction is the share of energy generated that came from PV. A site without a generator or a PV array has no figure for it.

	RENEWABLE FRACTION	FUEL PER KWH	PERFORMANCE RATIO	SITE LOSS
Dry River	46.2% • C	0.51 L/kWh • C	76.0% • C	*insufficient data* !
Granite Hill	100.0% • C	No generator	23.7% • C	16.4% • D
Palm Landing	50.9% • C	0.35 L/kWh • C	77.0% • C	5.0% • D
Tamarind Cross	100.0% • C	*not computable* ∅	71.0% • C	8.1% • D

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

4 Tariffs: an assumption

What customers were charged, and on what basis that is known.

The tariff is an assumption. Tariffs are assumptions, not current published rates. The records hold no tariff of their own: each tier's rate is an assumption used to price its bills, and it is not a published or approved rate. Each tier's billed revenue and energy sold are shown below; no rate is worked out from them here.

Billing by tariff tier, by site

As billed in the period. Energy sold to a prepaid account is energy purchased, not energy used.

	ACCOUNTS	ACTIVE	ENERGY SOLD (KWH)	REVENUE BILLED	REVENUE COLLECTED
Public institution Dry River	3 • C	3 • C	0 kWh • C	₦0 • C	₦0 • C
Tier 1 (basic household) Dry River	230 • C	230 • C	61 kWh • C	₦17,591 • C	₦17,591 • C
Tier 2 (household) Dry River	30 • C	30 • C	55 kWh • C	₦15,967 • C	₦15,967 • C
Tier 3 (commercial) Dry River	28 • C	28 • C	42 kWh • C	₦13,104 • C	₦13,104 • C
Tier 4 (productive use) Dry River	9 • C	9 • C	107 kWh • C	₦27,729 • C	₦27,729 • C
Public institution Granite Hill	2 • C	2 • C	0 kWh • C	₦0 • C	₦0 • C
Tier 1 (basic household) Granite Hill	110 • C	110 • C	21 kWh • C	₦5,403 • C	₦5,403 • C
Tier 2 (household) Granite Hill	8 • C	8 • C	6 kWh • C	₦1,539 • C	₦1,539 • C
Tier 3 (commercial) Granite Hill	9 • C	9 • C	0 kWh • C	₦0 • C	₦0 • C
Tier 4 (productive use) Granite Hill	1 • C	1 • C	0 kWh • C	₦0 • C	₦0 • C
Public institution Palm Landing	5 • C	5 • C	0 kWh • C	₦0 • C	₦9,134 • C
Tier 1 (basic household) Palm Landing	324 • C	300 • C	89 kWh • C	₦23,927 • C	₦23,927 • C
Tier 2 (household) Palm Landing	40 • C	40 • C	5 kWh • C	₦1,407 • C	₦1,407 • C
Tier 3 (commercial) Palm Landing	40 • C	40 • C	43 kWh • C	₦12,479 • C	₦12,479 • C
Tier 4 (productive use) Palm Landing	8 • C	8 • C	0 kWh • C	₦0 • C	₦40,573 • C
Public institution Tamarind Cross	6 • C	6 • C	0 kWh • C	₦0 • C	₦0 • C
Tier 1 (basic household) Tamarind Cross	330 • C	330 • C	118 kWh • C	₦30,724 • C	₦30,724 • C
Tier 2 (household) Tamarind Cross	60 • C	60 • C	14 kWh • C	₦3,744 • C	₦3,744 • C
Tier 3 (commercial) Tamarind Cross	55 • C	55 • C	75 kWh • C	₦20,941 • C	₦20,941 • C

	ACCOUNTS	ACTIVE	ENERGY SOLD (KWH)	REVENUE BILLED	REVENUE COLLECTED
Tier 4 (productive use) Tamarind Cross	14 • C	14 • C	111 kWh • C	₦25,468 • C	₦25,468 • C

Method: GridIntel reference collection (gridintel.atcc.reference 0.1.0).

5 Revenue and collection

What each site billed and what it collected.

Revenue and collection, by site

Cash basis: money received in the period against bills issued in the period.

	REVENUE BILLED	REVENUE COLLECTED	COLLECTION EFFICIENCY	REVENUE PER CONNECTION
Dry River	₦74,392 • C	₦74,392 • C	100.0% • C	₦247.97 • C
Granite Hill	₦6,942 • C	₦6,942 • C	100.0% • C	₦53.40 • C
Palm Landing	₦37,813 • C	₦87,520 • C	231.5% • C	₦96.22 • C
Tamarind Cross	₦80,877 • C	₦80,877 • C	100.0% • C	₦173.93 • C

Method: GridIntel reference collection (gridintel.atcc.reference 0.1.0); GridIntel reference site performance (gridintel.site.reference 0.1.0).

6 Each site's report against its records

Whether what each site reported agrees with what its own records show, figure by figure, on the report's basis.

How each report stands

A claim agrees when it is on the same basis as the calculation and within 1% of it.

	CLAIMS	AGREE	DIFFER	NOT COMPARABLE
Dry River	0 • C	0 • C	0 • C	0 • C
Granite Hill	0 • C	0 • C	0 • C	0 • C
Palm Landing	0 • C	0 • C	0 • C	0 • C
Tamarind Cross	0 • C	0 • C	0 • C	0 • C

Dry River

> No report from Dry River in the records states figures for 14 September 2026.

Granite Hill

> No report from Granite Hill in the records states figures for 14 September 2026.

Palm Landing

> No report from Palm Landing in the records states figures for 14 September 2026.

Tamarind Cross

> No report from Tamarind Cross in the records states figures for 14 September 2026.

7 Method, sources and how to reproduce this report

Where every figure came from, and how a reader can check that this document is the one the records give.

Methodologies

METHODOLOGY	VERSION	WHAT IT SAYS OF ITSELF
GridIntel reference collection `gridintel.atcc.reference`	`0.1.0`	GridIntel reference calculation — not a regulatory figure.
GridIntel reference site performance `gridintel.site.reference`	`0.1.0`	GridIntel reference calculation — not a regulatory figure.

Sources

- Synthetic mini-grid vending and billing (synthetic)
- Synthetic mini-grid meter data (synthetic)
- Synthetic mini-grid plant monitoring (synthetic)
- Synthetic mini-grid site register (synthetic)

> Every source above is synthetic: designed demonstration data, not records of a real organization.

How to reproduce it

1. Build the report with definition version 1, for 14 September 2026, from the Demonstration Mini-grid Operator (synthetic) records.
2. The data fingerprint must be `b0167ca08ac57457af3bea882205afb118f40ea88543b10d1bbe5fea52b64234`. SHA-256 over the sorted SHA-256 digests of every distinct result the report's services were given by the repositories, each written as JSON with its keys in order.
3. With the same definition, methodologies and fingerprint the report ID is the same: GI-MGC-2026-09-14-0B710C33. A different ID means something above differs.
4. The figures with their status, origin, unit and method are in the CSV of the same report ID: gridintel-compliance-2026-09-14-SAMPLE-SYNTHETIC.csv.

8 Regulatory references

> How far this report carries the fields of the instruments below, as read from the instruments themselves. It is not a claim that the report conforms to any regulator's or programme's template: none has been checked against the template it would be filed on.

INSTRUMENT	REQUIREMENT	THIS REPORT
NERC Mini-Grid Regulations 2023 (NERC-R-117-2023), s. 13(2) and s. 21(2)	Reports in the form prescribed in Schedule 4, per mini-grid or combined for a portfolio, at least once a year; M&E reports on the Schedule 4 template	**NOT CARRIED** Schedule 4 is not in the published text read (it ends at section 28). The report cannot be set against it until the template is supplied.
NERC Mini-Grid Regulations 2023 (NERC-R-117-2023), s. 16(1)	Supply in accordance with the contract with the community, as prescribed in Schedules 10 and 11	**PARTLY** Section 1 gives availability and hours of supply by site; the contracted service level is not in the records, so it is not tested.
NERC Mini-Grid Regulations 2023 (NERC-R-117-2023), s. 22(3)	Tariffs by the MYTO methodology; allowable technical losses not above 4% and non-technical losses not above 3%	**PARTLY** Section 4 states the tariff as an assumption. Section 3 gives site loss as a share of energy generated; it is not split into technical and non-technical loss.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Number of connections per site; customer acquisition	**PARTLY** Section 2 gives connections and active connections for the period; acquisition over time is not calculated.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Average consumption per user (ACPU)	**CARRIED** Section 2: consumption per connection (customer meters) and energy sold per connection (billing).
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Number of customers in each consumption bracket (<5 kWh, 5–10 kWh, ...)	**NOT CARRIED** Not calculated.

INSTRUMENT	REQUIREMENT	THIS REPORT
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Renewable energy fraction	**CARRIED** Section 3.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Percentage of time of available power supply at a site per month, as measured by inverter output	**PARTLY** Section 1 gives customer-weighted availability from interruption records, not from inverter output.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	SAIFI and SAIDI per site	**PARTLY** Section 1 gives interruptions per connection and lists every interruption with its duration; a site SAIDI is not stated as such.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Average revenue per user (ARPU); effective tariff	**PARTLY** Section 5 gives revenue per connection. No effective tariff is worked out: tariffs in this dataset are assumptions.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, PBG disbursement, milestone 2	Minimum kWh sold in a year = 30 kWh × connections achieved; every connection with consumption above 0 kWh	**NOT CARRIED** An annual test: not assessable from the period held, and connections with no consumption are not counted.

9 Glossary and key

Availability Customer-weighted share of the period with supply: 1 − (customer-hours interrupted ÷ customer-hours in the period). Every recorded interruption is counted.

Hours of supply a day The average hours a day the average connection had supply in the period.

Connection A customer connection in the site's register. An active connection is one billed in the period.

Energy sold Energy billed. For a prepaid account it is energy purchased, which may be used later.

Consumption Energy recorded by customer meters.

Renewable fraction The share of energy generated that came from PV.

Performance ratio PV energy delivered against what the array's rating and the measured irradiation would give; curtailment counts as a loss (IEC 61724).

Collection efficiency Revenue collected in the period ÷ revenue billed in the period, on a cash basis.

Claim A figure a site's own report states. It agrees when it is on the basis of the calculation and within the tolerance of it.

Tariff assumption A rate used to price bills in this dataset that is not a published or approved tariff.

Measured Read from a meter, a device or a source record.

Reported A figure published by someone else, shown as they stated it. It is never altered.

Calculated Computed by GridIntel from records, under a named and versioned methodology.

Estimated An estimate, not a measurement; the method is named with it.

Derived A residual: what is left when other figures are subtracted. It carries their uncertainty.

Insufficient data An input the figure needs is missing. No value is given rather than a guess, and never a zero.

Not available No source holds this figure, or the data does not cover the period.

Same basis A reported figure and a calculation count the same things over the same period by the same rule. Only figures on the same basis are compared.

Data fingerprint A SHA-256 digest of every record the report was built from. The same records give the same fingerprint; any change to them gives another.

Report ID The report's code, its period and the first characters of a digest of its definition, methodologies and data fingerprint. The same report of the same data has the same ID.

The marks beside a figure

• OK Every input is present and measured.

≈ Estimated inputs Calculated; every input is present, but at least one was estimated rather than measured.

! Insufficient data An input this figure needs is missing. No value is shown rather than a guess.

ø Not computable The inputs exist but the calculation is undefined for them.

– **Not available**	The platform has no source for this figure.
M **Measured**	Read from a meter, device or source record.
R **Reported**	Published by someone else; shown as stated.
C **Calculated**	Computed by GridIntel under a named methodology.
E **Estimated**	An estimate, not a measurement.
D **Derived**	A residual: what is left after subtracting other figures. It inherits their uncertainty.

SAMPLE: SYNTHETIC DATA. End of report GI-MGC-2026-09-14-0B710C33.